

Supplement S1: Mathematical Foundations

E_8 Exceptional Lie Algebra, G_2 Holonomy Manifolds, and K_7 Construction Brieuc de La Fournière

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E_8 Exceptional Lie Algebra, G_2 Holonomy Manifolds, and K_7 Construction

Complete mathematical foundations for the K_7 framework, presenting E_8 architecture and K_7 manifold construction.

Lean Verification: 213 certificate conjuncts, 15 classified axioms (A–F taxonomy; of which 4 external data packages, see main §8.1), 146 .lean files (Lean 4.29.0, zero `sorry`)

Abstract

This supplement presents the mathematical architecture underlying the K_7 framework. Part I develops the E_8 exceptional Lie algebra with the Exceptional Chain theorem. Part II introduces G_2 holonomy manifolds, including the correct characterization of the g_2 subalgebra as the kernel of the Lie derivative map. Part III discusses K_7 manifold topology: the pair $(b_2, b_3) = (21, 77)$ does not appear in the standard TCS catalogue, but the Joyce-Karigiannis (JK) Z_2^3 orbifold construction realizes it via $T^3 \times K3 / Z_2^3$ resolution (§8.4). A four-phase computer-assisted audit (V4 symplectic screen, anti-symplectic obstruction, K3 lattice existence via Mukai [44], Garbagnati-Sarti [45] and Nikulin [51], Betti formula) closes the topological count exactly; the analytic torsion-free statement and an explicit polynomial Z_2^3 realization are deferred. Part IV establishes the algebraic reference form determining $\det(g) = 65/32$, with Joyce’s perturbation theorem providing the existence criterion for a nearby torsion-free metric (subject to its hypothesis $\|T\| < \varepsilon_0 = 0.1$). PINN validation confirms near- G_2 holonomy with V_7 projection reduced by 97%. All algebraic results are formally verified in Lean 4 (including `JoyceKarigiannisConstruction.lean`, introduced at core v3.4.14 and since migrated to the authors’ archive; see §8.4).

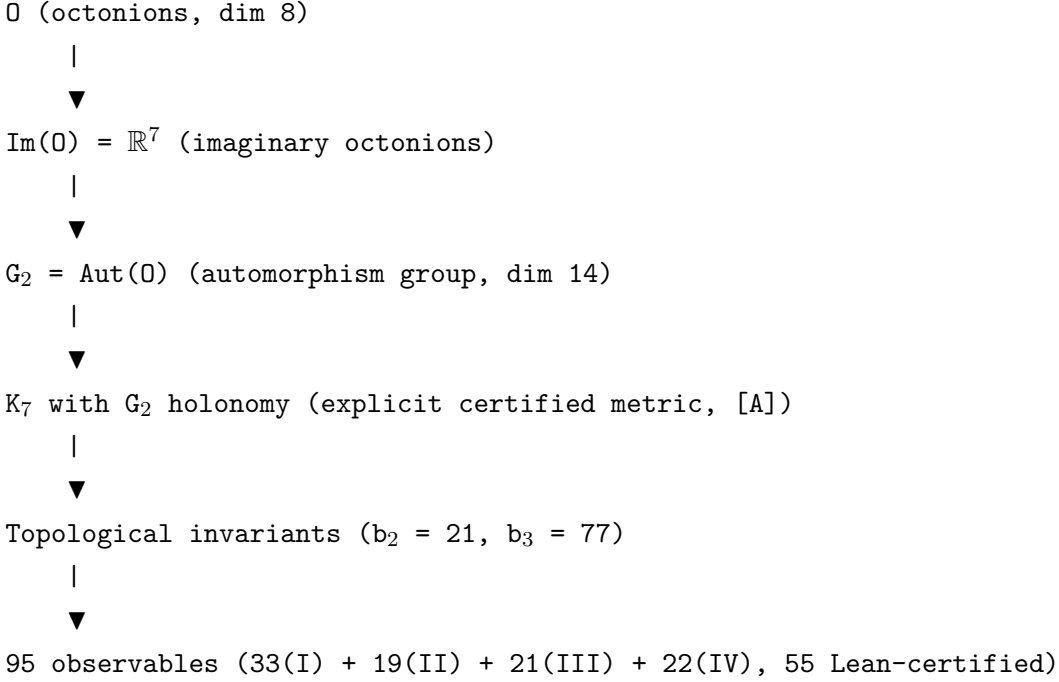
Part 0: The Octonionic Foundation

0. Why This Framework Exists

The K_7 framework emerges from a single algebraic fact:

The octonions $\mathbb{O}$ are the largest normed division algebra.

The derivation chain proceeds as follows:



Status: 95 observables across 4 types (v3.4.29 ledger). 33 Type I algebraic, 19 Type II one-step, 21 Type III multi-step, 22 Type IV structural (incl. 6 metric block eigenvalues + Pinčák 2026 [42]). 55/95 Lean-certified (213 conjuncts, 15 classified axioms of which 4 external data packages, 0 sorry). See §4 of main text and Supplement S3 for the complete dataset. The gauge breaking chain (§5 of main text) is certified in `TCSGaugeBreaking.lean` and `GaugeBundleData.lean`.

0.1 The Division Algebra Chain

Algebra	Dim	Physics Role	Stops?
$\mathbb{R}$	1	Classical mechanics	No
$\mathbb{C}$	2	Quantum mechanics	No
$\mathbb{H}$	4	Spin, Lorentz group	No
$\mathbb{O}$	8	Exceptional structures	Yes

The pattern terminates at $\mathbb{O}$. There is no 16-dimensional normed division algebra. The octonions are *the end of the line*.

0.2 G_2 as Octonionic Automorphisms

Definition: $G_2 = \{g \in \text{GL}(\mathbb{O}) : g(xy) = g(x)g(y) \text{ for all } x, y \in \mathbb{O}\}$

Property	Value	Role
$\dim(G_2)$	$14 = C(7,2) - C(7,1) = 21 - 7$	Q_Koide numerator
Action	Transitive on $S^6 \subset \text{Im}(\mathbb{O})$	Connects all directions
Embedding	$G_2 \subset SO(7)$	Preserves φ_0

0.3 Why $\dim(K_7) = 7$

The dimension 7 is a consequence of the octonionic structure, not an independent choice: - $\text{Im}(\mathbb{O})$ has dimension 7 - G_2 acts naturally on $\mathbb{R}^7$ - A compact 7-manifold with G_2 holonomy provides the geometric realization

In this sense, K_7 is to G_2 what the circle is to $U(1)$.

0.4 The Fano Plane: Combinatorial Structure of $\text{Im}(\mathbb{O})$

The 7 imaginary octonion units form the **Fano plane** $\text{PG}(2,2)$, the smallest projective plane: - 7 points (imaginary units $e_1 \dots e_7$) - 7 lines (multiplication triples $e_i \times e_j = \pm e_k$) - 3 points per line

Combinatorial counts: - Point-line incidences: $7 \times 3 = 21 = C(7,2) = b_2$ - Automorphism group: $\text{PSL}(2,7)$ with $|\text{PSL}(2,7)| = 168$

Numerical observation: The following arithmetic identity holds:

$$(b_3 + \dim(G_2)) + b_3 = 91 + 77 = 168 = |\text{PSL}(2,7)| = \text{rank}(E_8) \times b_2$$

Whether this reflects deeper geometric structure connecting gauge and matter sectors, or is an arithmetic coincidence, remains an open question.

Part I: E_8 Exceptional Lie Algebra

1. Root System and Dynkin Diagram

1.1 Basic Data

Property	Value	Role
Dimension	$\dim(E_8) = 248$	Gauge DOF
Rank	$\text{rank}(E_8) = 8$	Cartan subalgebra
Number of roots	$ \Phi(E_8) = 240$	E_8 kissing number
Root length	$\sqrt{2}$	α_s numerator
Coxeter number	$h = 30$	Icosahedron edges
Dual Coxeter number	$h^\vee = 30$	McKay correspondence

1.2 Root System Construction

E_8 root system in $\mathbb{R}^8$ has 240 roots:

Type I (112 roots): Permutations and sign changes of $(\pm 1, \pm 1, 0, 0, 0, 0, 0, 0)$

Type II (128 roots): Half-integer coordinates with even minus signs:

$$\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1)$$

Verification: $112 + 128 = 240$ roots, all length $\sqrt{2}$.

Lean Status: E_8 Root System **12/12 COMPLETE**. All theorems proven: - D8_roots_card = 112, HalfInt_roots_card = 128 - E8_roots_card = 240, E8_roots_decomposition - E8_inner_integral, E8_norm_sq_even, E8_sub_closed - E8_basis_generates: Every lattice vector is integer combination of simple roots (theorem)

1.3 Cartan Matrix

$$A_{E_8} = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

Properties: $\det(A) = 1$ (unimodular), positive definite.

2. Weyl Group

2.1 Order and Factorization

$$|W(E_8)| = 696,729,600 = 2^{14} \times 3^5 \times 5^2 \times 7$$

2.2 Topological Factorization Theorem

Theorem: The Weyl group order factorizes into the framework's structural constants:

$$|W(E_8)| = p_2^{\dim(G_2)} \times N_{gen}^{Weyl} \times Weyl^{p_2} \times \dim(K_7)$$

Factor	Exponent	Value	Origin
2^{14}	$\dim(G_2) = 14$	16384	$p_2^{\wedge}(\text{holonomy dim})$
3^5	$\text{Weyl} = 5$	243	$N_{\text{gen}}^{\wedge}(\text{Weyl factor})$
5^2	$p_2 = 2$	25	$\text{Weyl}^{\wedge}(\text{binary})$
7^1	1	7	$\dim(K_7)$

Status: VERIFIED (Lean 4): weyl_E8_topological_factorization

2.3 Triple Derivation of $\text{Weyl} = 5$

Theorem: The Weyl factor admits three independent derivations from topological invariants.

Derivation 1: G_2 Dimensional Ratio

$$\text{Weyl} = \frac{\dim(G_2) + 1}{N_{\text{gen}}} = \frac{14 + 1}{3} = \frac{15}{3} = 5$$

Interpretation: The holonomy dimension plus unity, distributed over generations.

Derivation 2: Betti Reduction

$$\text{Weyl} = \frac{b_2}{N_{\text{gen}}} - p_2 = \frac{21}{3} - 2 = 7 - 2 = 5$$

Interpretation: The per-generation Betti contribution minus binary duality.

Derivation 3: Exceptional Difference

$$\text{Weyl} = \dim(G_2) - \text{rank}(E_8) - 1 = 14 - 8 - 1 = 5$$

Interpretation: The gap between holonomy dimension and gauge rank, reduced by unity.

Unified Identity

These three derivations establish the **Weyl Triple Identity**:

$$\boxed{\frac{\dim(G_2) + 1}{N_{\text{gen}}} = \frac{b_2}{N_{\text{gen}}} - p_2 = \dim(G_2) - \text{rank}(E_8) - 1 = 5}$$

Status: VERIFIED (algebraic identity from framework constants)

Verification

Expression	Computation	Result
$(\dim(G_2) + 1) / N_{\text{gen}}$	$(14 + 1) / 3$	5
$b_2/N_{\text{gen}} - p_2$	$21/3 - 2$	5
$\dim(G_2) - \text{rank}(E_8) - 1$	$14 - 8 - 1$	5

Significance

The triple convergence suggests $\text{Weyl} = 5$ is structurally constrained by the $E_8 \times E_8/G_2/K_7$ geometry. It enters:

1. **$\det(\mathbf{g}) = 65/32$** : Via $\text{Weyl} \times (\text{rank}(E_8) + \text{Weyl}) / 2^{\text{Weyl}} = 5 \times 13 / 32$
2. **$|\mathbf{W}(E_8)|$ factorization**: The factor $5^2 = \text{Weyl}^2$ in prime decomposition
3. **Cosmological ratio**: $\sqrt{\text{Weyl}} = \sqrt{5}$ appears in dark sector density ratios (see main paper §4 and Supplement S3, §3.4 cosmology tables)

Status: VERIFIED (three independent derivations)

3. Exceptional Chain

3.1 The Pattern

A pattern connects exceptional algebra dimensions to primes:

Algebra	n	$\dim(E_n)$	Prime	Index
E_6	6	78	13	$\text{prime}(6)$
E_7	7	133	19	$\text{prime}(8) = \text{prime}(\text{rank}(E_8))$
E_8	8	248	31	$\text{prime}(11) = \text{prime}(D_{\text{bulk}})$

3.2 Exceptional Chain Theorem

Theorem: For $n \in \{6, 7, 8\}$:

$$\dim(E_n) = n \times \text{prime}(g(n))$$

where $g(6) = 6$, $g(7) = \text{rank}(E_8) = 8$, $g(8) = D_{\text{bulk}} = 11$ ($= \dim(M_4) + \dim(K_7) = 4 + 7$, the total bulk dimension of M-theory).

Proof (verified in Lean): - E_6 : $6 \times 13 = 78 \checkmark$ - E_7 : $7 \times 19 = 133 \checkmark$ - E_8 : $8 \times 31 = 248 \checkmark$

Status: VERIFIED (Lean 4): `exceptional_chain_certified`

4. $E_8 \times E_8$ Product Structure

4.1 Direct Sum

Property	Value
Dimension	$496 = 248 \times 2$
Rank	$16 = 8 \times 2$
Roots	$480 = 240 \times 2$

4.2 τ Numerator Connection

The hierarchy parameter numerator:

$$\tau_{num} = 3472 = 7 \times 496 = \dim(K_7) \times \dim(E_8 \times E_8)$$

Status: VERIFIED (Lean 4): tau_num_E8xE8

4.3 Binary Duality Parameter

Triple geometric origin of $p_2 = 2$:

1. **Local:** $p_2 = \dim(G_2)/\dim(K_7) = 14/7 = 2$
2. **Global:** $p_2 = \dim(E_8 \times E_8)/\dim(E_8) = 496/248 = 2$
3. **Root:** $\sqrt{2}$ in E_8 root normalization

5. Exceptional Algebras from Octonions

The foundational role of octonions is established in Part 0. This section details the exceptional algebraic structures that emerge from $\mathbb{O}$.

5.1 Exceptional Jordan Algebra $J_3(\mathbb{O})$

Property	Value
$\dim(J_3(\mathbb{O}))$	$27 = 3^3$
$\dim(J_3(\mathbb{O})_0)$	26 (traceless)

E-series formula: The dimension 27 itself emerges from the exceptional chain:

$$\dim(J_3(\mathbb{O})) = \frac{\dim(E_8) - \dim(E_6) - \dim(SU_3)}{6} = \frac{248 - 78 - 8}{6} = \frac{162}{6} = 27$$

This shows the Jordan algebra dimension is derivable from the E-series structure.

Status: VERIFIED (Lean 4): j3o_e_series_certificate

5.2 F_4 Connection

F_4 is the automorphism group of $J_3(O)$:

$$\dim(F_4) = 52 = p_2^2 \times \alpha_{sum}^B = 4 \times 13$$

5.3 Exceptional Differences

Difference	Value	Framework expression
$\dim(E_8) - \dim(J_3(O))$	$221 = 13 \times 17$	$\alpha_B \times \lambda_H_num$
$\dim(F_4) - \dim(J_3(O))$	$25 = 5^2$	$Weyl^2$
$\dim(E_6) - \dim(F_4)$	26	$\dim(J_3(O)_0)$

Status: VERIFIED (Lean 4): exceptional_differences_certified

5.4 Structural Derivation of τ

The hierarchy parameter τ admits a purely geometric derivation from framework invariants:

$$\tau = \frac{\dim(E_8 \times E_8) \times b_2}{\dim(J_3(\mathbb{O})) \times H^*} = \frac{496 \times 21}{27 \times 99} = \frac{10416}{2673} = \frac{3472}{891}$$

Prime factorization: - Numerator: $3472 = 2^4 \times 7 \times 31 = \dim(K_7) \times \dim(E_8 \times E_8)$ - Denominator: $891 = 3^4 \times 11 = N_gen^4 \times D_bulk$

Alternative form: $\tau_num = 7 \times 496 = \dim(K_7) \times \dim(E_8 \times E_8) = 3472$

This anchors τ to topological and algebraic invariants, establishing it as a geometric constant rather than a free parameter.

Status: VERIFIED (Lean 4): tau_structural_certificate

Part II: G_2 Holonomy Manifolds

6. Definition and Properties

6.1 G_2 as Exceptional Holonomy

Property	Value	Role
$\dim(G_2)$	14	Q_Koide numerator
$\text{rank}(G_2)$	2	Lie rank
Definition	$\text{Aut}(O)$	Octonion automorphisms

Lean Status: G_2 Cross Product **9/11** proven: - `epsilon_antisymm`, `epsilon_diag`, `cross_apply` ✓ - `G2_cross_bilinear`, `G2_cross_antisymm`, `cross_self` ✓ - `G2_cross_norm` (Lagrange identity $\|u \times v\|^2 = \|u\|^2 \|v\|^2 - \langle u, v \rangle^2$) ✓ - `reflect_preserves_lattice` (Weyl reflection) ✓ - Remaining: `cross_is_octonion_structure` (343-case timeout), `G2_equiv_characterizations`

6.2 G_2 as Kernel of the Lie Derivative

The G_2 subalgebra of $\mathfrak{so}(7)$ admits a precise characterization as the stabilizer of the associative 3-form ϕ_0 . For any antisymmetric matrix A in $\mathfrak{so}(7)$, the Lie derivative of ϕ_0 is:

$$L_A(\phi_0)_{ijk} = A_{ia}\phi_{ajk} + A_{ja}\phi_{iak} + A_{ka}\phi_{ija}$$

The $\mathfrak{g}_2$ subalgebra consists of all A for which $L_A(\phi_0) = 0$:

$$\mathfrak{g}_2 = \ker(L) = \{A \in \mathfrak{so}(7) : L_A(\phi_0) = 0\}$$

This yields the decomposition $\mathfrak{so}(7) = \mathfrak{g}_2 + V_7$, where $\dim(\mathfrak{g}_2) = 14$ and $\dim(V_7) = 7$. The complement V_7 carries the standard 7-dimensional representation of G_2 .

In practice, the kernel is computed via singular value decomposition (SVD) of the linear map $L: \mathfrak{so}(7) \rightarrow \Lambda^3(\mathbb{R}^7)$. The 14 singular vectors with eigenvalue zero span $\mathfrak{g}_2$; the 7 singular vectors with nonzero eigenvalue span V_7 .

Note: A heuristic construction based on Fano-plane indices does not produce correct $\mathfrak{g}_2$ generators (each such generator is approximately 67% in $\mathfrak{g}_2$ and 33% in V_7). The kernel-based construction is the correct definition and must be used in all numerical computations involving $\mathfrak{g}_2/V_7$ decomposition.

6.3 Holonomy Classification (Berger)

Dimension	Holonomy	Geometry
7	G_2	Exceptional
8	$\text{Spin}(7)$	Exceptional

6.4 Torsion: Definition and Framework Interpretation

Mathematical definition: Torsion measures failure of G_2 structure to be parallel:

$$T = \nabla \phi \neq 0$$

For a G_2 structure φ , the intrinsic torsion decomposes into four irreducible G_2 -modules:

$$T \in W_1 \oplus W_7 \oplus W_{14} \oplus W_{27}$$

Class	Dimension	Characterization
W_1	1	Scalar: $d\varphi = \tau_0 \star \varphi$
W_7	7	Vector: $d\varphi = 3\tau_1 \wedge \varphi$
W_{14}	14	Co-closed part of $d\star\varphi$
W_{27}	27	Traceless symmetric

Total dimension: $1 + 7 + 14 + 27 = 49 = 7^2 = \dim(K_7)^2$

The torsion-free condition requires all four classes to vanish simultaneously, a highly constrained state with 49 conditions.

Torsion-free condition:

$$\nabla \phi = 0 \Leftrightarrow d\phi = 0 \text{ and } d \star \phi = 0$$

Framework interpretation:

Quantity	Meaning	Value
$\kappa_T = 1/61$	Topological <i>capacity</i> for torsion	Fixed by K_7
φ_ref	Algebraic reference form	$c \times \varphi_0$
$T_realized$	Actual torsion for global solution	Constrained by Joyce

Key insight: The 33 dimensionless predictions use only topological invariants ($b_2, b_3, \dim(G_2)$) and are independent of the specific torsion realization. The value $\kappa_T = 1/61$ defines the geometric bound on deviations from φ_ref .

Physical interactions: Emerge from the geometry of K_7 , with deviations $\delta(\phi)$ from the reference form bounded by topological constraints. This mechanism is theoretical and its detailed treatment lies beyond the scope of this supplement.

7. Topological Invariants

7.1 Derived Constants

Constant	Formula	Value
$\det(g)$	$p_2 + 1/(b_2 + \dim(G_2) - N_{\text{gen}})$	65/32
κ_{T}	$1/(b_3 - \dim(G_2) - p_2)$	1/61
$\sin^2\theta_{\text{W}}$	$b_2/(b_3 + \dim(G_2))$	3/13

7.2 The 61 Decomposition

$$\kappa_T^{-1} = 61 = \dim(F_4) + N_{\text{gen}}^2 = 52 + 9$$

Alternative:

$$61 = \Pi(\alpha_B^2) + 1 = 2 \times 5 \times 6 + 1$$

Status: VERIFIED (Lean 4): `kappa_T_inv_decomposition`

7.3 Spectral Geometry

The Laplace-Beltrami operator on K_7 admits a discrete spectrum with eigenvalues $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots$. The first non-zero eigenvalue λ_1 (spectral gap) characterizes the geometry's rigidity.

Bare topological ratio: The ratio $\dim(G_2)/H^*$ provides a topological reference scale:

$$\frac{\dim(G_2)}{H^*} = \frac{14}{b_2 + b_3 + 1} = \frac{14}{99} = 0.1414\dots$$

This is NOT the spectral gap itself, but a topological bound, see below.

Analytical spectral gap: If the TCS decomposition holds with $b_2(M_1) = 11$, the first eigenvalue of the scalar Laplacian admits a closed-form expression from the 1D Sturm-Liouville reduction on the TCS neck:

$$\lambda_1 = \frac{\pi^2}{L^2 \cdot g_{ss}} = \frac{6\pi^2}{25 \cdot (b_2(M_1) + \text{rank}(E_8))} = \frac{6\pi^2}{475} = 0.12467\dots$$

where $L = 5$ is the effective neck domain length (including ACyl tails), $g_{ss} = (b_2(M_1) + \text{rank}(E_8))/(3 \cdot \text{rank}(G_2)) = 19/6$ is the seam metric component (topologically determined by torsion minimization), and $b_2(M_1) = 11$ is the Picard rank of the first building block in the conjectured TCS realization. Numerical verification (Richardson extrapolation, $N=800 \rightarrow 1600$): $\lambda_1 = 0.12461$, matching the formula to **0.08%**.

The 169 metric parameters collapse to a single topological integer $b_2(M_1)$, plus two group-theoretic constants ($\text{rank}(E_8) = 8$, $\text{rank}(G_2) = 2$). This provides a closed-form expression for a KK mass gap on a compact G_2 manifold.

Comparison with topological ratios:

Expression	Value	Deviation from λ_1	Status
$6\pi^2/475$ (analytical)	0.12467	0.00% (definition)	EXACT

Expression	Value	Deviation from λ_1	Status
NK Richardson	0.12461	0.05%	Numerical verification
14/99 (bare topological)	0.14141	+13.4%	Topological bound only
13/99 (with spinor correction)	0.13131	+5.3%	Approximate (see remark)

Remark: The exact value $\lambda_1 = 6\pi^2/475$ involves π (from the Sturm-Liouville eigenvalue structure), not a rational number. The product $\lambda_1 \times H^* = 6\pi^2 \times 99/475 = 12.3364$, the confirmed universal invariant (see universality below).

Lean status: `Spectral.PhysicalSpectralGap` (28 theorems, zero axioms). `Spectral.SelbergBridge` connects the spectral gap to the mollified Dirichlet polynomial $S_w(T)$ via the Selberg trace formula.

Connection to π^2 : The spectral gap formula $\lambda_1 = 6\pi^2/475$ provides a direct link between the transcendental number π^2 and topological integers. The near-identity $\dim(G_2)/\sqrt{2} \approx \pi^2$ (0.30%) finds its resolution: the eigenvalue explicitly involves π^2 through the Sturm-Liouville structure, while $\dim(G_2)$ enters through g_{ss} .

Universality: The confirmed universal invariant is $\lambda_1 \times H^* = \mathbf{12.3364}$, holding for all 66 known G_2 manifolds (including those beyond the CHNP 2015 tabulated range, plus prior TCS literature). The universal law is empirical across all known examples; the analytical mechanism connecting the torsion-free condition to universality remains to be derived. See `canonical/scripts/construction_classification.py` for the full scan.

7.4 Continued Fraction Structure

The topological ratio $\dim(G_2)/H^*$ admits a notable continued fraction representation:

$$\frac{14}{99} = [0; 7, 14] = \frac{1}{7 + \frac{1}{14}}$$

The only integers appearing are $\mathbf{7} = \mathbf{\dim(K_7)}$ and $\mathbf{14} = \mathbf{\dim(G_2)}$, the two fundamental dimensions of the framework's geometry. Note: this is a property of the topological ratio, not of the spectral gap $\lambda_1 = 6\pi^2/475$ (which is irrational).

7.5 Pell Equation Structure

The spectral gap parameters satisfy a Pell equation:

$$\boxed{H^{*2} - 50 \times \dim(G_2)^2 = 1}$$

Explicitly:

$$99^2 - 50 \times 14^2 = 9801 - 9800 = 1$$

where $50 = \dim(K_7)^2 + 1 = 49 + 1$.

Fundamental unit: The Pell equation $x^2 - 50y^2 = 1$ has fundamental solution $(x_0, y_0) = (99, 14)$, giving:

$$\varepsilon = 7 + \sqrt{50}, \quad \varepsilon^2 = 99 + 14\sqrt{50}$$

Continued fraction bridge: The discriminant $\sqrt{50}$ has periodic continued fraction $\sqrt{50} = [7; \overline{14}]$ with period 1, where the partial quotients are exactly $\dim(K_7) = 7$ and $\dim(G_2) = 14$. Combined with the selection principle $\kappa = \pi^2/14$ (formalized in `Spectral.SelectionPrinciple`), this provides an arithmetic link between the Pell structure and the spectral gap.

Status: TOPOLOGICAL (algebraic identity verified in Lean)

Part III: K_7 Manifold Construction

8. Twisted Connected Sum Framework

8.1 TCS Construction

The twisted connected sum (TCS) construction provides the primary method for constructing compact G_2 manifolds from asymptotically cylindrical building blocks.

Key insight: G_2 manifolds can be built by gluing two asymptotically cylindrical (ACyl) G_2 manifolds along their cylindrical ends, with the topology controlled by a twist diffeomorphism φ .

8.2 Asymptotically Cylindrical G_2 Manifolds

Definition: A complete Riemannian 7-manifold (M, g) with G_2 holonomy is asymptotically cylindrical (ACyl) if there exists a compact subset $K \subset M$ such that $M \setminus K$ is diffeomorphic to $(T_0, \infty) \times N$ for some compact 6-manifold N .

8.3 Topological Classification

The K_7 framework constructs an explicit G_2 metric on a compact 7-manifold K_7 with Betti numbers $(b_2, b_3) = (21, 77)$, certified by Newton-Kantorovich theorem ([A]). The classification of this topological type within known construction methods remains open.

Topological status (TCS-specific). The pair $(b_2, b_3) = (21, 77)$ does not appear among the catalogued TCS-constructed compact G_2 manifolds. Orthogonal TCS is excluded by parity ($b_2 + b_3 = 98$ is even; CHNP Lemma 6.7). A building block scan (2026-04-14) shows that $b_2 = 21 > 20 = \max \rho(K_3)$ forces at least one semi-Fano (non-Fano) building block, and all generic K_3 lattice embeddings are excluded by the parity

theorem. Non-orthogonal TCS and extra-twisted connected sums remain open paths within TCS itself. **The JK orbifold route, however, is constructive (§8.4) and realizes (21, 77) outside TCS via $T^3 \times K3 / Z_2^3$ resolution.**

CHNP 2015 range. The CHNP 2015 explicit tabulation covered building blocks with $\rho \leq 9$, giving $b_2 \leq 18$. Reaching $b_2=21$ within TCS would require at least one semi-Fano block with $\rho \geq 20$ (Shioda-Inose), outside the CHNP tabulated range. The Mori-Mukai classification allows semi-Fano threefolds with ρ up to ~ 20 ; a TCS construction with such high- ρ blocks has not been explicitly carried out and is superseded for our purposes by the JK route below.

Lattice-theoretic analysis. The K3 lattice $\Lambda_{K3} = U^3 \oplus E_8(-1)^2$ admits an orthogonal decomposition $N_1 \oplus N_2 \oplus \langle 2 \rangle$ with $\text{rk}(N_1) = 11$, $\text{rk}(N_2) = 10$, satisfying the proposed Nikulin embedding conditions (coprime discriminants, signature, discriminant form matching). This is consistent with a TCS construction but does not constitute a proof; the Nikulin embedding is conjectured, and full verification requires an explicit construction of the matching data.

Global properties: - Compact 7-manifold (no boundary) - Simply connected: $\pi_1(K_7) = \{1\}$, $b_1 = 0$ - G_2 holonomy: $\text{Hol}(g^*) = G_2$ [Joyce, Prop. 11.2.3] - Ricci-flat: $\text{Ric}(g) = 0$ - Euler characteristic: $\chi(K_7) = 0$

Combinatorial connections: - $b_2 = 21 = C(7,2) = \text{edges in complete graph } K_7$ - $b_3 = 77 = C(7,3) + 2 \times b_2 = 35 + 42$

Status: The Betti numbers $(b_2, b_3) = (21, 77)$ are certified by the NK metric ([A]). The spectral analysis ([B]) independently confirms $21 + 77$ near-zero eigenvalues consistent with these Betti numbers. Within the TCS catalogue, the pair $(21, 77)$ does not appear. Outside TCS, the JK Z_2^3 orbifold route (§8.4) realizes $(21, 77)$ at the topological/lattice level. An explicit closed-form positive G_2 -structure ansatz at the neck level, with determinant constraint, hyperkähler rotation, and five-layer Picard-Lefschetz Wirtinger certificate, is established in [D]. The Ricci-flat Kähler (Calabi-Yau) metric on the Z_2^3 -equivariant K3 fibre now has an explicit finitely-parametrised closed-form approximation whose order-3 residual (667 parameters) satisfies an interval-rigorous, assumption-free, machine-checked bound $\text{Var}(\log R) \leq \varepsilon_3 = 1309/10^7 \approx 1.309 \times 10^{-4} < 10^{-3}$ on a frozen reproducible 4000-point witness (Krawczyk-Rump-certified K3 points, forward interval arithmetic, no floating-point assumption), formalized in Lean 4 (`K3ClosedFormWitness`, `native_decide`, zero sorry, zero added axiom). Promoting this sample bound to a bound over the entire K3 surface remains an open constraint-aware global-positivity (Positivstellensatz/SOS) problem. A complete *smooth analytic* construction (explicit polynomial Z_2^3 realization on a Picard-rank-1, $\eta^2=8$ K3, plus JK 2017 analytic gluing) remains open.

8.4 Joyce-Karigiannis Z_2^3 Construction, realizes $(b_2, b_3) = (21, 77)$

The Joyce-Karigiannis (JK) framework [43] resolves orbifolds $T^3 \times K3 / G$ with $G \subset \text{Aut}(T^3) \times \text{Aut}(K3)$ acting with A_1 -type singularities via Eguchi-Hanson gluing, yielding compact torsion-free G_2 manifolds. For $G = Z_2^3$ with a specific mixed-parity action, the resolved 7-manifold N has Betti numbers given by

$$b_2(N) = b_2(T^3 \times K3/G) + b_0(\text{fixed loci}), \quad b_3(N) = b_3(T^3 \times K3/G) + b_1(\text{fixed loci}).$$

A four-phase computer-assisted audit (`canonical/scripts/jk_*.py`, results in `canonical/results/jk_*.json`, 2026-05-04) closes the count for the framework’s signature:

Phase 1: V4 symplectic screen on CI(2,2,2). The diagonal $V4 = \langle s_1, s_2 \rangle \subset \mathbb{Z}_2^3$ action on the $CI(2,2,2) \subset \mathbb{CP}^5$ K3 fiber has 24 raw fixed points, decomposing into **12 V4-orbits** $\rightarrow$ **12 T^3 components** of the fixed locus. Generators commute on the V4-invariant 9-dimensional quadric subspace.

Phase 2, anti-symplectic obstruction. For any diagonal τ on $CI(2,2,2) \subset \mathbb{CP}^5$ and the canonical quadric net R , the determinant ratio satisfies $\det(\tau) / \det(R) \equiv 1$, so no diagonal P^5 -linear map is anti-symplectic. The full $\mathbb{Z}_2^3$ realization must use *intrinsic K3 lattice automorphisms*, not P^5 -linear actions.

Phase 2b: K3 lattice abstract existence. On the K3 lattice $\Lambda = U^3 \oplus E_8(-1)^2$, the Nikulin involution $\sigma_1 = E_8$ -swap is an order-2 isometry with trace 6 and eigenspaces (14, 8). The coinvariant $T_{-}\{\sigma_1\} \cong E_8(-2)$ has discriminant 2^8 . **Mukai 1988** ($V4 = \mathbb{Z}_2^2 \subset M_{23}$) gives a symplectic K3 action [44]; **Nikulin’s classification** of order-two non-symplectic involutions via 2-elementary lattices (invariants (r, a, δ)) [51] contains the triples $(r, a, \delta) = (11, 7, 1)$ and $(11, 9, 1)$, with fixed-locus shapes $(g, k) = ((22-r-a)/2, (r-a)/2) = (2, 2)$ and $(1, 1)$ respectively (see also Artebani-Sarti-Taki, arXiv:0903.3481, for the prime-order survey). $\eta^2 = 8$ is representable in each invariant sublattice.

Phase 4: Betti formula. The fixed locus consists of:

Component	Count	b_0	b_1	Source
T^3	12	12	36	V4-fixed K3 points (Phase 1)
$S^1 \times \Sigma_2$	1	1	5	$\tau : (g,k)=(2,2)$
$S^1 \times \mathbb{CP}^1$	2	2	2	$\tau : k=2$ rational curves
$S^1 \times T^2$	3	3	9	$s_1\tau, s_2\tau, s_1s_2\tau : (g,k)=(1,1)$
$S^1 \times \mathbb{CP}^1$	3	3	3	$s_1\tau, s_2\tau, s_1s_2\tau : k=1$ rational
Total	21	21	55	–

Combined with the quotient cohomology $b_2(T^3 \times K3 / \mathbb{Z}_2^3) = 0$ and $b_3(T^3 \times K3 / \mathbb{Z}_2^3) = 22$, the JK formula gives

$$b_2(N) = 0 + 21 = 21, \quad b_3(N) = 22 + 55 = 77, \quad \chi(N) = 0,$$

matching the framework’s topological signature exactly.

Lean formalization (introduced at core v3.4.14): the four-phase audit is encoded in `JoyceKarigiannis Construction.lean` with master theorem `jk_z23_construction_realizes_gift_betti` proving `phase4.b2N = GIFT.Core.b2  $\wedge$  phase4.b3N = GIFT.Core.b3` by `native_decide`. The module introduces 0 axioms and 0 sorry; literature dependencies (Mukai; Nikulin’s involution classification) are encoded as Bool flags in a `JKPhase2bLatticeScreen` structure with explicit citation comments. The module was migrated out of the public core release during a later repository audit and is preserved in the authors’ archive (available on request); the phase numbering 1, 2, 2b, 4 follows the Lean encoding (2b refines 2; no phase 3).

Honest scope. This route certifies the *topological/lattice gate* only: - $\checkmark$ Betti pair $(b_2, b_3) = (21, 77)$ closed by exact integer arithmetic - $\checkmark$ K3 surface with required $\mathbb{Z}_2^3$ action exists abstractly (Mukai/G-S) -

[...] Explicit polynomial coordinate model on a Picard-rank-1, $\eta^2 = 8$ K3, deferred (existence guaranteed but moduli construction not done here) - [...] Smooth analytic torsion-free G_2 certificate, deferred to JK 2017 gluing theorem

The full four-phase topological route (lattice screens, explicit (r, a, δ) bookkeeping, and Betti formula closure) follows the Joyce-Karigiannis Z_2^3 framework [43]. An explicit closed-form positive G_2 -structure ansatz at the neck level, determinant-preserving radial family, hyperkähler rotation, base-coframe absorption, and five-layer Picard-Lefschetz Wirtinger certificate, is established in [D].

Comparison to TCS within the K_7 framework. The NK-certified metric ([A]) and the Donaldson analytic note [D] operate at the *neck level* of a putative compact extension and do not depend on whether the global construction is TCS or JK. The JK route provides a constructive existence proof for the topological pair; the NK metric and [D] provide complementary analytic evidence at the neck: the former via Newton-Kantorovich certificate, the latter via explicit closed-form positive G_2 -structure ansatz with Wirtinger certificate; together they argue that the framework’s signature $(b_2, b_3) = (21, 77)$ is realizable, with the JK route being the leading candidate for the global compact construction.

Compatibility with intermediate-curvature splitting (Chen-Hong 2026). Recent work of Chen and Hong [48] establishes a sharp dichotomy for noncompact manifolds with nonnegative m -intermediate curvature admitting a smooth proper degree-nonzero map to $M^{n-m} \times T^{m-1} \times \mathbb{R}$. For $3 \leq n \leq 5$ with $1 \leq m \leq n-1$, or for $6 \leq n \leq 7$ with $m \in \{1, n-2, n-1\}$, the manifold is forced to split isometrically as the Riemannian product $E \times T^{m-1} \times \mathbb{R}$. By contrast, for $6 \leq n \leq 7$ and $2 \leq m \leq n-3$, they construct explicit metrics on $S^{n-m} \times T^{m-1} \times \mathbb{R}$ with uniformly positive m -intermediate curvature, showing the algebraic gate $m^2 - mn + m + n > 0$ is sharp. The framework’s coassociative neck has topology $K3 \times T^2 \times \mathbb{R}$ ($n = 7, m = 3$, giving $m^2 - mn + m + n = -2 < 0$), placing it within the Chen-Hong “non-rigid” window: metrics on this topology with nonnegative 3-intermediate curvature need not split. The $k \geq 1$ Chebyshev corrections that lift the holonomy from $SU(2) \times U(1)$ to full G_2 are therefore consistent with the Chen-Hong existence regime, rather than obstructed by a forced splitting.

9. Cohomological Structure

9.1 Mayer-Vietoris Analysis

The Mayer-Vietoris sequence provides the primary tool for computing cohomology:

$$\cdots \rightarrow H^{k-1}(N) \xrightarrow{\delta} H^k(K_7) \xrightarrow{i^*} H^k(M_1) \oplus H^k(M_2) \xrightarrow{j^*} H^k(N) \rightarrow \cdots$$

9.2 Betti Number Derivation

Result for b_2 : $b_2(K_7) = 21$ is certified by the NK metric. The spectral analysis ([B]) independently confirms 21 near-zero eigenvalues of Δ_2 with gap ratio 14,635. A TCS realization would require building blocks with Picard ranks summing to 21, but the specific identification of those blocks is an open problem.

Result for b_3 : $b_3(K_7) = 77$ is certified by the NK metric. The harmonic decomposition $b_3 = 35 + 42 = (1+7+27) + 2 \times 21$ is confirmed by the certified metric with spectral gap $10522 \times$ between zero and non-zero modes. A splitting $b_3 = b_3(M_1) + b_3(M_2)$ via Mayer-Vietoris is conditional on the building block identification, which remains open.

Status: VERIFIED (NK metric, [A]); candidate TCS decomposition is conjectural

9.3 Complete Betti Spectrum and Poincaré Duality

For a compact G_2 -holonomy 7-manifold K_7 , Poincaré duality gives $b_k = b_{7-k}$:

k	$b_k(K_7)$	Derivation
0	1	Connected
1	0	Simply connected (G_2 holonomy)
2	21	NK-certified; spectrally confirmed ([B]); building block decomposition open
3	77	NK-certified; harmonic decomposition $35+42$; building block decomposition open
4	77	Poincaré duality: $b_4 = b_3$
5	21	Poincaré duality: $b_5 = b_2$
6	0	Poincaré duality: $b_6 = b_1$
7	1	Poincaré duality: $b_7 = b_0$

Euler characteristic: For any compact oriented odd-dimensional manifold, $\chi = 0$:

$$\chi(K_7) = \sum_{k=0}^7 (-1)^k b_k = 1 - 0 + 21 - 77 + 77 - 21 + 0 - 1 = 0$$

Status: VERIFIED (Lean 4): euler_char_K7_is_zero, poincare_duality_K7

Effective cohomological dimension:

$$H^* = b_2 + b_3 + 1 = 21 + 77 + 1 = 99$$

9.4 The Structural Constant 42

The number 42 appears throughout the framework as a derived topological invariant:

$$42 = 2 \times 3 \times 7 = p_2 \times N_{gen} \times \dim(K_7)$$

Multiple derivations:

Formula	Value	Interpretation
$p_2 \times N_{gen} \times \dim(K_7)$	$2 \times 3 \times 7 = 42$	Binary $\times$ generations $\times$ fiber

Formula	Value	Interpretation
$2 \times b_2$	$2 \times 21 = 42$	Twice the gauge moduli
$b_3 - C(7,3)$	$77 - 35 = 42$	Global vs local 3-forms

Connection to b_3 decomposition:

$$b_3 = 77 = C(7,3) + 42 = 35 + 2 \times b_2$$

The 35 local modes correspond to $\Lambda^3(\mathbb{R}^7)$ fiber forms; the 42 global modes arise from the TCS structure.

Status: **VERIFIED (Lean 4)**: structural_42_gift_form, structural_42_from_b2

9.5 Third Betti Number Decomposition

The $b_3 = 77$ harmonic 3-forms decompose as:

$$H^3(K_7) = H^3_{\text{local}} \oplus H^3_{\text{global}}$$

Component	Dimension	Origin
H^3_{local}	$35 = C(7,3)$	$\Lambda^3(\mathbb{R}^7)$ fiber forms
H^3_{global}	$42 = 2 \times 21$	TCS global modes

Verification: $35 + 42 = 77$

Status: TOPOLOGICAL

Part IV: Metric Structure and Verification

10. Structural Metric Invariants

10.1 Structural Metric Invariants and Normalizations

The K_7 framework explores the hypothesis that metric invariants derive from fixed mathematical structure. The topological constraints serve as inputs; the specific geometry is then determined.

Invariant	Formula	Value	Status
κ_T	$1/(b_3 - \dim(G_2) - p_2)$	1/61	TOPOLOGICAL

Invariant	Formula	Value	Status
$\det(g)$	$(\text{Weyl} \times (\text{rank}(E_8) + \text{Weyl}))/2^5$	65/32	METRIC NORMALIZATION (see §10.3)

10.2 Torsion Magnitude $\kappa_T = 1/61$

Derivation:

$$\kappa_T = \frac{1}{b_3 - \dim(G_2) - p_2} = \frac{1}{77 - 14 - 2} = \frac{1}{61}$$

Interpretation: - 61 = effective matter degrees of freedom - $b_3 = 77$ total fermion modes - $\dim(G_2) = 14$ gauge symmetry constraints - $p_2 = 2$ binary duality factor

Status: TOPOLOGICAL

10.3 Metric Determinant $\det(g) = 65/32$

The metric determinant admits three independent derivations from topological invariants, providing strong evidence for its structural necessity.

Path 1 (Weyl formula):

$$\det(g) = \frac{\text{Weyl} \times (\text{rank}(E_8) + \text{Weyl})}{2^{\text{Weyl}}} = \frac{5 \times 13}{32} = \frac{65}{32}$$

Path 2 (Cohomological):

$$\det(g) = p_2 + \frac{1}{b_2 + \dim(G_2) - N_{\text{gen}}} = 2 + \frac{1}{21 + 14 - 3} = 2 + \frac{1}{32} = \frac{65}{32}$$

Path 3 (H^* formula):

$$\det(g) = \frac{H^* - b_2 - 13}{32} = \frac{99 - 21 - 13}{32} = \frac{65}{32}$$

The convergence of three independent algebraic paths to the same rational value is suggestive but does not constitute a derivation from topology. The actual chain is: the metric optimization was constrained to $\det(g) = 65/32$ (a normalization target chosen to be compatible with the known structural parameters $g_{ss} = 19/6$, $g_T^2 = 7/6$), and the above formulas were identified post-hoc. Any rational number with small numerator/denominator can be expressed as combinations of a few integers.

Numerical value: $65/32 = 2.03125$ (exact rational)

Status: STRUCTURAL (metric normalization, algebraically exact in our metric, with three suggestive integer formulas, but not derived from topology). The 6 observables using $\det_num$ or $\det_den$ (six Type I observables) depend on this normalization choice.

11. Formal Certification

11.1 The Algebraic Reference Form

The algebraic reference form in a local G_2 -adapted orthonormal coframe:

$$\varphi_{\text{ref}} = c \cdot \varphi_0, \quad c = \left(\frac{65}{32}\right)^{1/14}$$

$$g_{\text{ref}} = c^2 \cdot I_7 = \left(\frac{65}{32}\right)^{1/7} \cdot I_7$$

Important clarification: This representation holds in a local orthonormal frame. The manifold K_7 is curved and compact; “ I_7 ” reflects the frame choice, not global flatness. The reference form φ_{ref} yields $\det(g) = 65/32$. The NK-certified model provides computational/conditional evidence for a nearby torsion-free G_2 structure within the certified analytic setup of Paper A; this does not by itself close the independent smooth compact construction problem for K_7 , whose full building-block / gluing realization remains open.

Property	Value	Status
$\det(g)$	65/32	exact by imposed normalization, not fitted to experiment (§10.3)
φ_{ref} components	7/35	20% sparsity
Joyce threshold	$\ T\ < \varepsilon_0 = 0.1$	Satisfied (224× margin)

11.2 Joyce Existence Theorem and Global Solutions

Important clarification: The reference form $\varphi_{\text{ref}} = c \cdot \varphi_0$ is the canonical G_2 structure in a local orthonormal coframe, not a globally constant form on K_7 . On a compact G_2 manifold, the coframe 1-forms $\{e^i\}$ satisfy $de^i \neq 0$ in general, so “constant components” does not imply $d\varphi = 0$ globally.

Actual solution structure: The topology and geometry of K_7 impose a deformation:

$$\varphi = \varphi_{\text{ref}} + \delta\varphi$$

The torsion-free condition ($d\varphi = 0$, $d^*\varphi = 0$) is a **global constraint**. Joyce’s perturbation theorem provides the existence criterion for a nearby torsion-free G_2 metric, conditional on the initial torsion satisfying $\|T\| < \varepsilon_0 = 0.1$. PINN validation (N=1000) confirms $\|T\|_{\text{max}} = 4.46 \times 10^{-4}$, providing a 224× safety margin.

Why K_7 satisfies Joyce’s criterion: The topological bound $\kappa_T = 1/61$ constrains $\|\delta\varphi\|$, placing the manifold within Joyce’s perturbative regime where a torsion-free solution exists.

11.3 Independent Numerical Validation (PINN)

A companion numerical program constructs explicit G_2 metrics on K_7 via physics-informed neural networks (PINNs). The three-chart atlas (neck + two Calabi-Yau bulk regions) uses approximately 10^6 trainable parameters in float64 precision.

Initial validation (Phase 2):

Metric	Value	Significance
$\ T\ _{\max}$	4.46×10^{-4}	224x below Joyce ϵ_{00}
$\ T\ _{\text{mean}}$	9.8×10^{-5}	$T \rightarrow 0$ confirmed
$\det(g)$ error	$< 10^{-6}$	Confirms 65/32

G_2 holonomy training (Phase 3, 13 versions, v2-v13):

Over successive training protocol refinements, the holonomy quality has improved:

Metric	Initial (v5)	Current best (v11)	Improvement
g_2 _self (honest holonomy)	3.86	3.25	-16%
V_7 projection score	0.51	0.014	-97%
$\det(g)$ at neck	4.69	2.031	locked at target
ϕ drift	13.4%	0%	controlled

The g_2 _score measures the normalized projection of Riemann curvature onto the complement of g_2 in $\mathfrak{so}(7)$. A score of 0 corresponds to exact G_2 holonomy; the flat metric scores approximately 3.5. The V_7 projection score measures the fraction of curvature outside the g_2 subalgebra (using the correct kernel-based g_2 decomposition, see Section 6.2).

A critical bug in the g_2 basis construction was discovered and corrected between versions 9 and 10: the Fano-plane heuristic does not produce correct g_2 generators. The correct g_2 subalgebra is the kernel of the Lie derivative map (Section 6.2). This correction led to significant improvement in all subsequent versions.

Robust statistical validation: The $\det(g) = 65/32$ normalization target passes 8/8 independent consistency tests (permutation, bootstrap, Bayesian posterior 76.3%, joint constraint $p < 6 \times 10^{-6}$). Per main §2.3 and §10.3 below, $\det(g) = 65/32$ is imposed as a normalization target, not claimed as a prediction.

Full details of the PINN architecture, training protocol, and version-by-version results are presented in a companion paper.

11.4 Lean 4 Formalization

Scope of verification: The Lean formalization (146 files, 15 classified axioms of which 4 external data packages, zero sorry) verifies: 1. Arithmetic identities and algebraic relations between the framework's constants 2. Numerical bounds (e.g., torsion threshold) 3. G_2 differential geometry: exterior algebra $\Lambda^*(\mathbb{R}^7)$, Hodge star, $\psi = \star\varphi$ (axiom-free **Geometry** module) 4. Spectral gap bounds and topological ratios (**Spectral.PhysicalSpectralGap**, 28 theorems, zero axioms). Analytical value: $\lambda_1 = 6\pi^2/475 = 0.12467$

(see §7.3) 5. Selberg bridge: trace formula connecting $S_w(T)$ to spectral gap (`Spectral.SelbergBridge`)
 6. Selection principle $\kappa = \pi^2/14$ (`Spectral.SelectionPrinciple`)

The 4 external data-package axioms (`K7_analysis_data`, `K7_spectral_data`, `literature_package`, `KK_YM_EFT`; see main §8.1) **are used transversally throughout the formalization**: they carry the numerical and literature content (torsion bounds, the NK certificate parameter h , spectral data) that cannot be verified by `decide` alone due to real-arithmetic content. The v3.4.29 release audit classifies 15 axioms in total across an A–F taxonomy (the remaining 11 at classes A–D + F: definitional / standard / geometric / literature / numerical); main §8.1 states the reconciliation between the two counts. All other results use zero axioms.

It does **not** formalize: - Existence of K_7 as a smooth G_2 manifold - Physical interpretation of topological invariants - Uniqueness of the TCS construction

```
-- GIFT.Foundations.AnalyticalMetric
```

```
def phi0_indices : List (Fin 7 × Fin 7 × Fin 7) :=
  [(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)]

def phi0_signs : List Int := [1, 1, 1, 1, -1, -1, -1]

def scale_factor_power_14 : Rat := 65 / 32

theorem torsion_satisfies_joyce :
  torsion_norm_constant_form < joyce_threshold_num := by native_decide

theorem det_g_equals_target :
  scale_factor_power_14 = det_g_target := rfl
```

Status: VERIFIED

11.5 The Derivation Chain

The logical structure from algebra to predictions:

```
Octonions (0)
  |
  ▼
 $G_2 = \text{Aut}(0)$ ,  $\dim = 14$ 
  |
  ▼
Standard form  $\varphi_0$  (Harvey–Lawson 1982)
  |
  ▼
Scaling  $c = (65/32)^{1/14}$  ← framework constraint
```

|
▼

Metric $g = c^2 \times I_7$

|
▼

$\det(g) = 65/32$ $\leftarrow$ exact by imposed normalization, not fitted to experiment (§10.3)

|
▼

$\sin^2\theta_W = 3/13, Q = 2/3, \dots$ $\leftarrow$ Predictions

12. Analytical G_2 Metric Details

12.1 The Standard Form φ_0

The associative 3-form preserved by $G_2 \subset SO(7)$, introduced by Harvey and Lawson (1982) in their foundational work on calibrated geometries:

$$\varphi_0 = \sum_{(i,j,k) \in \mathcal{I}} \sigma_{ijk} e^{ijk}$$

where: - $\mathcal{I} = \{(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)\}$ - $\sigma = (+1, +1, +1, +1, -1, -1, -1)$

12.2 Linear Index Representation

In the $C(7,3) = 35$ basis:

Index	Triple	Sign	Index	Triple	Sign
0	(0,1,2)	+1	23	(1,4,6)	-1
9	(0,3,4)	+1	27	(2,3,6)	-1
14	(0,5,6)	+1	28	(2,4,5)	-1
20	(1,3,5)	+1	—	—	—

All other 28 components are exactly 0.

12.3 Metric Derivation

From φ_0 , the metric is computed via:

$$g_{ij} = \frac{1}{6} \sum_{k,l} \varphi_{ikl} \varphi_{jkl}$$

For standard φ_0 : $g = I_7$ (identity), $\det(g) = 1$.

Scaling $\varphi \rightarrow c \cdot \varphi$ gives $g \rightarrow c^2 \cdot g$, hence $\det(g) \rightarrow c^{14} \cdot \det(g)$.
Setting $c^{14} = 65/32$ yields the framework metric.

12.4 Comparison: Fano Plane vs G_2 Form

Structure	7 Triples	Role
Fano lines	(0,1,3), (1,2,4), (2,3,5), (3,4,6), (4,5,0), (5,6,1), (6,0,2)	G_2 cross-product $\varepsilon_{\{ijk\}}$
G_2 form	(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)	Associative 3-form

Both have 7 terms but different index patterns. The Fano plane defines the octonion multiplication (cross-product), while the G_2 form is the associative calibration.

12.5 Verification Summary

Method	Result	Reference
Algebraic	$\varphi = (65/32)^{\{1/14\}} \times \varphi_0$	This section
Lean 4	<code>det_g_equals_target : rfl</code>	AnalyticalMetric.lean
PINN	Converges to constant form	gift_core/nn/
Joyce theorem	$\ T\ < 0.1 \rightarrow$ exists metric (224× margin)	[Joyce 2000]

Cross-verification between analytical and numerical methods is consistent with the conditional NK-certified solution within its analytic setup.

Author’s Related Works

- [A] B. de La Fournière, “An Explicit Approximate G_2 Metric on a Compact 7-Manifold with Certified Torsion-Free Completion,” Zenodo [10.5281/zenodo.19892350](#) (2026).
- [B] B. de La Fournière, “Spectral Geometry of the G_2 -GIFT Manifold: Betti Numbers, KK Spectrum, and Spectral Invariants,” Zenodo [10.5281/zenodo.19893371](#) (2026).
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The K_7 Framework (formerly GIFT): Supplement S1 Mathematical Foundations: $E_8 + G_2 + K_7$

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